

Person-Level Kinship Verification: Leakage-Free Evaluation, Set-Based Aggregation, and a Negative Result for Quantum-Inspired Metrics

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Abstract—Facial kinship verification is evaluated almost exclusively as a photograph-versus-photograph decision on four benchmarks. We argue that this formulation is both harder to measure reliably and weaker than the data supports. First, we quantify the leakage admitted by conventional splitting: because Families In the Wild draws 57,120 pairs from 95 families and 472 identities, a random pair-level split places an identity from *every* test pair into the training set, a violation we measure at 11,424/11,424 (100%). We propose a leakage-free protocol based on grouped k -fold cross-validation in which every family is tested exactly once and negatives are regenerated within each fold; this reduces seed-to-seed accuracy variation from 22.5 to 12.1 points. Second, we show that the conventional formulation discards most of the available evidence: FIW supplies a median of five photographs per identity, yet the standard protocol scores one against one. Representing each *person* by their photograph set, as template-based face recognition does, improves ROC-AUC by +0.077 on FIW under the leakage-free protocol, while reducing exactly to the single-image case on corpora storing one photograph per person. We further show that scoring father–mother–child triads jointly, rather than as independent pairs, adds +0.032 ROC-AUC on TSKinFace ($p = 0.0001$). Finally we report a systematic negative result: nine quantum-inspired formulations, spanning similarity measures, feature extractors and inductive biases, each evaluated against a capacity-matched classical control, fail to improve on their controls, and four are significantly worse. All protocol and evaluation code is released.

Index Terms—Facial kinship verification, evaluation protocol, dataset bias, data leakage, quantum-inspired machine learning, negative results.

I. INTRODUCTION

Facial kinship verification asks whether two facial images depict biologically related individuals. Progress is measured on a small set of public benchmarks, and reported accuracies have risen

steadily. This paper examines whether the evaluation protocols supporting those numbers are sound.

We make three contributions, of which the first two concern how the task is posed and measured rather than which model is applied to it.

A leakage-free protocol, and the leakage it removes. Splitting a corpus at the level of *pairs* rather than *identities* does not isolate a test set. FIW draws 57,120 pairs from only 95 families and 472 identities, so identities recur across pairs an average of 242 times; under a random pair-level split we measure that 100% of test pairs share an identity with training. Correcting this is less direct than it appears, because every negative pair joins two families and, treating families as vertices, yields a graph with a single connected component over all of FIW: no family-disjoint partition of the existing pairs exists, and negatives must be regenerated within each fold. That kinship datasets carry construction artefacts is known — prior work has shown that negatives assembled by recombining family members admit shortcuts [1] — but we are not aware of a quantification of identity leakage under the splitting practice in common use, nor of a released protocol that removes it.

Person-level rather than photograph-level verification. The conventional formulation compares two photographs. FIW supplies a median of five photographs per identity, so this discards most of the evidence about each person. Template-based protocols in face recognition score a person represented by an image set [2]; to our knowledge this has not been applied to kinship verification. Doing so improves ROC-AUC by +0.077 on FIW, and by construction leaves corpora with one photograph per person unchanged. We additionally show that father–mother–child triads carry information that pairwise decomposition destroys, worth a further +0.032 ROC-AUC.

A negative result for quantum-inspired metrics. Quantum-inspired similarity has been proposed for this task. We evaluate nine formulations against capacity-matched classical controls; none improves on its control and four are significantly worse. We give a structural account of why, rather than attributing the outcome to implementation.

Concretely we contribute: (i) a measured quantification of identity leakage under conventional splitting, and a released protocol that removes it; (ii) person-level and triadic formulations of the task, with gains measured under that protocol; and (iii) a pre-registered nine-formulation negative result for quantum-inspired metrics.

II. RELATED WORK

A. Kinship verification

Early work framed the task as metric learning, learning a projection under which related faces lie closer than unrelated ones [3]. Subsequent methods replaced the hand-designed metric with deep pairwise encoders and, more recently, attention- and graph-based architectures [4]. Evaluation is concentrated on four corpora: KinFaceW-I and KinFaceW-II [5], TSKinFace [6], and Families In the Wild [7], the last also underpinning the RFIW challenge series [8].

Reported accuracies are obtained under each dataset’s official protocol. We therefore do not tabulate them against our results: Section III shows that those protocols admit two forms of optimistic bias, so a direct ranking would mislead in both directions.

B. Evaluation bias in vision benchmarks

That benchmark protocols can inflate reported performance is established in adjacent fields. Face recognition distinguishes subject-disjoint from random splitting precisely because identity overlap leaks information [9], and template-based protocols such as IJB-B/C score a *person* represented by several images rather than a single photograph [2]. More generally, models exploit incidental correlations in preference to the intended signal, a failure mode catalogued as shortcut learning [10] and, earlier, as dataset bias [11].

Both concerns apply directly to kinship verification. Construction artefacts in kinship corpora have been reported [1], but we are not aware of a quantification of identity leakage under the

TABLE I
IDENTITY LEAKAGE UNDER A RANDOM 80/20 PAIR-LEVEL SPLIT OF FIW.

Quantity	Value
Pairs	57,120
Distinct families	95
Distinct identities	472
Mean appearances per identity	242
Largest family, share of all pairs	19%
Test pairs sharing an identity with train	11,424 / 11,424 (100%)
Test pairs whose family was seen in train	11,424 / 11,424 (100%)

splitting practice in common use, nor of a person-level protocol for this task: existing multi-image work addresses image-set subspace representations or matching within family photographs, which is a different problem from aggregating a subject’s photographs into a single template for pairwise kinship decisions [12]. Sections III and VII address both gaps.

C. Quantum-inspired machine learning

Quantum feature maps embed classical data in a Hilbert space where inner products act as kernels [13], with reported gains on classical tasks [14]. Related lines use SWAP-test state fidelity as a similarity measure and density matrices as classifiers. Such methods have been proposed for kinship verification, and the claim that they outperform classical counterparts on classical data is contested; dequantization results show that several purported advantages vanish under scrutiny [15].

Section VIII contributes to that debate empirically, evaluating nine formulations against capacity-matched classical controls.

III. SOURCES OF OPTIMISTIC BIAS

A. Identity leakage under pair-level splitting

Let $\mathcal{P}$ be a set of labelled pairs and $\text{id}(\cdot)$ the identity depicted in an image. A split $(\mathcal{P}_{\text{tr}}, \mathcal{P}_{\text{te}})$ is *identity-disjoint* if

$$\left(\bigcup_{p \in \mathcal{P}_{\text{tr}}} \text{id}(p) \right) \cap \left(\bigcup_{p \in \mathcal{P}_{\text{te}}} \text{id}(p) \right) = \emptyset. \quad (1)$$

Random splitting over $\mathcal{P}$ does not enforce this. Table I reports the measured violation on FIW.

B. A non-obvious obstacle to correcting it

Splitting on families is not sufficient by itself. Each *negative* pair joins two families, and treating families as vertices and negatives as edges yields a graph over FIW with a single connected component containing all 95 families. No family-disjoint partition of the existing pairs exists. Negatives must therefore be discarded and regenerated *within* each side after the split. We found this empirically: a splitter that passed synthetic unit tests failed on the real corpus for exactly this reason.

A second consequence is data loss. Because pre-existing negatives straddle any group boundary, naively dropping them removes usable data — on KinFaceW-I, 173 of 1,066 pairs, leaving only 119 test pairs. Regenerating negatives per side instead recovers the test set to 212 pairs on KinFaceW-I and 400 on KinFaceW-II.

C. Family membership as a predictor in TSKinFace

That kinship benchmarks carry construction artefacts is established. Prior work has shown that negatives assembled by recombining members of family photographs permit a model to succeed without learning kinship [1], and the TSKinFace protocol documents its own negative-construction procedure [6]. We do not claim to be the first to observe that this dataset family is vulnerable.

The specific regularity we measure is distinct from the same-photograph effect, and worth separating. TSKinFace stores one cropped face per family member, so *no* pair — positive or negative — shares a source photograph, and the same-photograph shortcut cannot arise. What does hold is that all 800 positives are drawn within a family and all 800 negatives across families, so family membership predicts the label perfectly. A model can exploit correlated capture conditions — illumination, camera, background — shared by members photographed together.

To measure the margin this affords, we rebuild the negatives so that half are drawn *within* a family, pairing father against mother: co-photographed, hence sharing every session cue, but not kin.

The 0.0398 ROC-AUC reduction is measured on raw embeddings without training, so it reflects separability available to any model rather than a property of ours. All TSKinFace results in this paper use the shortcut-free construction.

TABLE II
TSKINFACE PAIR CONSTRUCTION. UNDER THE STANDARD PROTOCOL FAMILY MEMBERSHIP PREDICTS THE LABEL PERFECTLY; NO PAIR SHARES A SOURCE PHOTOGRAPH IN EITHER CONSTRUCTION.

Construction	Same-fam. pos.	Same-fam. neg.	Cosine AUC
Standard	100%	0%	0.7278
Shortcut-free (ours)	100%	50%	0.6880

We check separately that our FIW results do not rest on the same-photograph effect that prior work identifies: 8.2% of FIW positives are drawn from a single source photograph (against 0% of negatives), and excluding them changes accuracy by +0.08 points (82.53% $\rightarrow$ 82.61%). The effect is present in FIW but our model does not exploit it.

IV. CORRECTED PROTOCOL

We adopt grouped k -fold cross-validation with $k = 5$. The grouping unit is the family, which is the unit that shares identities and capture conditions.

- 1) **Positives only are partitioned.** Families are dealt largest-first into the currently smallest fold, which keeps folds balanced despite the highly skewed family sizes.
- 2) **Negatives are generated per side**, after partitioning, from the families present on that side.
- 3) **Group keys are dataset-scoped.** The family identifier `fd_003` occurs in both KinFaceW-I and KinFaceW-II denoting different families; an unscoped key merges them and corrupts the disjointness check.
- 4) **Family contribution is capped** at 1,500 pairs, preventing the largest FIW family (19% of all pairs) from dominating whichever fold it enters.
- 5) **Thresholds are calibrated on a group-disjoint validation slice** of the training side. The test fold is scored once.

Fold integrity is verified directly rather than assumed: zero folds exhibit train/test group overlap, and every family is tested exactly once with no duplicates (95/95 FIW, 533/533 KinFaceW-I, 1000/1000 KinFaceW-II, 559/559 TSKinFace).

A. Effect on measurement stability

Table III shows why a single group-disjoint split is insufficient. Accuracy on one split varies by up

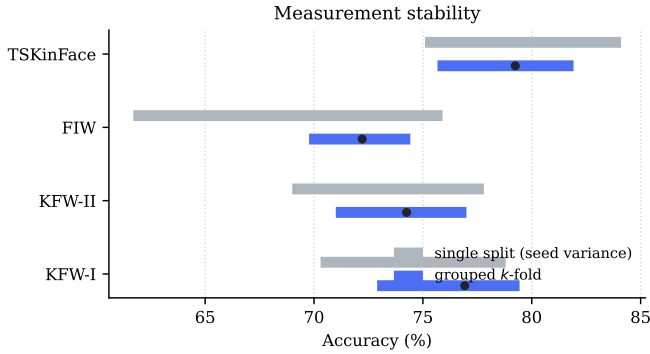


Fig. 1. Accuracy range under a single group-disjoint split across random seeds (grey) against grouped k -fold (blue, mean marked). A single split is too unstable to support a headline number: on FIW the seed-to-seed range spans 14.2 points.

TABLE III
ACCURACY RANGE ACROSS SEEDS (SINGLE SPLIT) VERSUS
ACROSS FOLDS (GROUPED k -FOLD).

Dataset	Single split	Grouped k -fold
FIW	61.7–75.9 (14.2)	69.8–74.4 (4.6)
KinFaceW-I	70.3–78.8 (8.5)	72.9–79.4 (6.5)
KinFaceW-II	69.0–77.8 (8.8)	71.0–77.0 (6.0)
TSKinFace	75.1–84.1 (9.0)	75.7–81.9 (6.2)
Overall spread	22.5	12.1

to 14.2 points with the random seed, because FIW’s test side reduces to a small number of families of very unequal size.

V. REFERENCE MODEL

To measure the protocol rather than an architecture, we use a deliberately conventional model. Frozen FaceNet embeddings (512-d) pass through a shared Siamese encoder; kinship is symmetric, so only symmetric pair features are formed — sum, absolute difference, elementwise product, and cosine — and concatenated with a one-hot relation code before a three-layer MLP. Predictions are averaged over both argument orders, making the model exactly symmetric. The model has 497,881 parameters.

Its plainness is deliberate: findings in Sections III and IV cannot then be attributed to architectural novelty. We note that a linear probe on the same frozen embeddings reaches 0.702 ROC-AUC against this model’s 0.810 on FIW, confirming the head performs non-trivial work and that the frozen backbone is not the binding constraint.

TABLE IV
GROUPED 5-FOLD RESULTS. EVERY FAMILY TESTED EXACTLY
ONCE.

Dataset	Accuracy (95% CI)	ROC-AUC
KinFaceW-I	76.92 ± 2.19	0.8752 ± 0.0146
KinFaceW-II	74.25 ± 1.95	0.8320 ± 0.0190
FIW	72.21 ± 1.55	0.8101 ± 0.0175
TSKinFace (shortcut-free)	79.25 ± 2.32	0.8814 ± 0.0192
Mean	75.66	0.8497

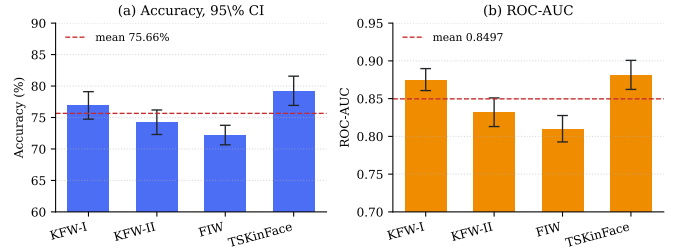


Fig. 2. Grouped 5-fold results, every family tested exactly once. (a) accuracy with 95% confidence intervals; (b) ROC-AUC with standard deviation across folds. Dashed line marks the four-dataset mean.

VI. RESULTS

Table IV reports results under the corrected protocol. We deliberately omit a comparison against published accuracies. Those numbers are obtained under each dataset’s official protocol, which permits the leakage and shortcut quantified in Section III; presenting a ranking against them would mislead in both directions. We regard establishing comparable baselines under the corrected protocol as work the community must do collectively, and we release our implementation to that end.

A. Per-relation analysis

Performance is markedly uneven across relation types. On FIW, father–daughter reaches 0.9933 ROC-AUC while mother–son reaches 0.7124 — and mother–son is the largest subset (5,296 pairs). Averaged over datasets the ordering is mother–son (0.8288), mother–daughter (0.8973), father–son (0.9172), father–daughter (0.9312). We verified that this is not a labelling artefact: gender fields in the FIW metadata are complete and well-formed across the families we audited. Cross-gender parent–child verification appears intrinsically harder, and we regard it as the most promising target for future architectural work.

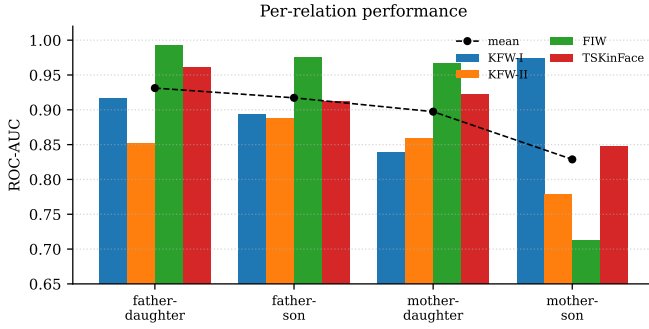


Fig. 3. Per-relation ROC-AUC. Mother-son is the weakest relation on three of four datasets and, on FIW, the largest subset (5,296 pairs).

TABLE V

SET-LEVEL REPRESENTATION UNDER GROUPED 5-FOLD. GAINS APPEAR ONLY WHERE PHOTO SETS EXIST; SINGLE-PHOTOGRAPH CORPORA ARE BIT-IDENTICAL TO THE BASELINE, AS REQUIRED.

Dataset	Set size	Single	Set-level	Δ
KinFaceW-I	1.00	0.7210	0.7210	0.0000
KinFaceW-II	1.00	0.7672	0.7672	0.0000
TSKinFace	1.00	0.7306	0.7306	0.0000
FIW	7.30	0.7296	0.8066	+0.0771

VII. SET-LEVEL AND TRIADIC REPRESENTATIONS

Sections III–VI concern how the task is *measured*. We now examine two ways in which the task is conventionally *posed* that discard available evidence.

A. Identity photo sets

The standard protocol scores one photograph against one photograph. FIW, however, supplies a median of five photographs per identity, so the conventional formulation discards most of the evidence for each person. We represent each identity by its photo set and form symmetric set statistics: the ℓ_2 -normalised centroid, the within-set spread (1 minus mean pairwise cosine), the set size, and the maximum, minimum, mean and standard deviation of the cross-set similarity matrix.

A set of size one must reduce exactly to the single-image case: the centroid is the embedding, the spread is zero, and every cross-set statistic collapses to the single cosine. We assert this in the test suite rather than assume it, which matters because the three remaining benchmarks store exactly one photograph per person.

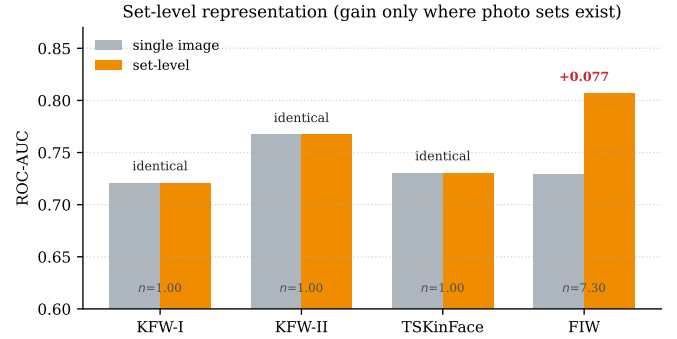


Fig. 4. Set-level representation. The gain appears only where photo sets exist (n = mean images per identity). The three single-photograph corpora are bit-identical to the single-image baseline, which is the degradation guarantee asserted in the test suite.

Table V reports the outcome. On FIW the representation is worth +0.0771 ROC-AUC, closely matching an untrained embedding-level estimate of +0.071, which indicates the gain is intrinsic to the representation rather than a product of additional model capacity. The three single-photograph corpora are unchanged to machine precision.

We stress the narrowness of this result: it is a large gain on one benchmark and exactly zero on three, and any aggregate over the four would obscure that. The practical implication is that kinship verification deployments in which several photographs of a person are available — the common case — are leaving a substantial margin unclaimed, and that benchmarks storing a single photograph per identity cannot measure this.

B. Triadic structure

TSKinFace supplies father, mother and child jointly, but the pairwise protocol decomposes each triad into two independent comparisons. We instead score the triad: alongside $\cos(F, C)$ and $\cos(M, C)$ we compute the best convex parental mixture $\max_{\alpha} \cos(\alpha F + (1 - \alpha)M, C)$, the maximising α , the gain of that mixture over the better single parent, and the parental similarity $\cos(F, M)$.

The last quantity is not expressible in a pairwise formulation and is the strongest single contributor: a child resembling one parent is weaker evidence of kinship when the two parents already resemble each other.

Triadic scoring improves ROC-AUC by +0.0324 over the best single parent (paired t -test over folds, $t = 14.93$, $p = 0.0001$; the gain is positive in all

TABLE VI
TRIADIC SCORING ON TSKinFACE, GROUPED 5-FOLD, FOLDS
ASSIGNED BY THE PARENTS' FAMILY.

Arm	Accuracy (95% CI)	ROC-AUC
Best single parent	67.62 ± 2.69	0.7569
Triadic	69.05 ± 2.65	0.7893
Triadic + phase sweep	69.50 ± 2.41	0.7887

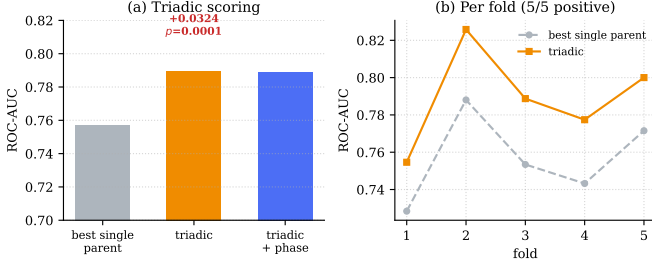


Fig. 5. Triadic scoring on TSKinFace. (a) ROC-AUC by arm; (b) per fold, with the triadic arm ahead in all five.

five folds). Negatives retain the true parents and substitute a child from another family, so the label depends on kinship rather than on any property of the parent pair.

VIII. A NEGATIVE RESULT FOR QUANTUM-INSPIRED METRICS

A. Formulations tested

We evaluated nine formulations, chosen to span the three roles a quantum component can occupy: a similarity measure, a feature extractor, and an inductive bias. Each is compared against a control matched for capacity or intent (Table VII).

Under grouped k -fold with 20 paired observations, enabling the SWAP-test branch yields 75.66% against 76.34% with it disabled (accuracy $p = 0.158$; ROC-AUC $p = 0.982$). The branch also costs 6–14 $\times$ runtime.

B. A cautionary observation

The purity regulariser initially measured +1.3 ROC-AUC on a single seed, with reduced variance — a result consistent with effective regularisation. Repeated across nine paired runs the sign reversed, and the method proved significantly *worse* than no regulariser (-0.0089 , $p = 0.047$). We report this because the single-seed reading was of the kind that is readily published, and it illustrates why capacity-matched controls and repeated trials are necessary when evaluating claims of this type.

TABLE VII
QUANTUM-INSPIRED FORMULATIONS AGAINST MATCHED CLASSICAL CONTROLS.

Formulation	Control	Outcome
SWAP fidelity (5 seeds)	classical head	$p = 0.945$
SWAP fidelity (20 folds)	classical head	$p = 0.982$
Density fidelity $\text{Tr}(\rho_1 \rho_2)$	mean-pooling	-8.7 AUC
Interference phase sweep	classical mixture	$+0.1$ AUC
Entanglement entropy	cosine	0.514 alone
POVM head	capacity-matched	-5.5 AUC
Purity regulariser	decorrelation	-0.9 , $p = 0.047$

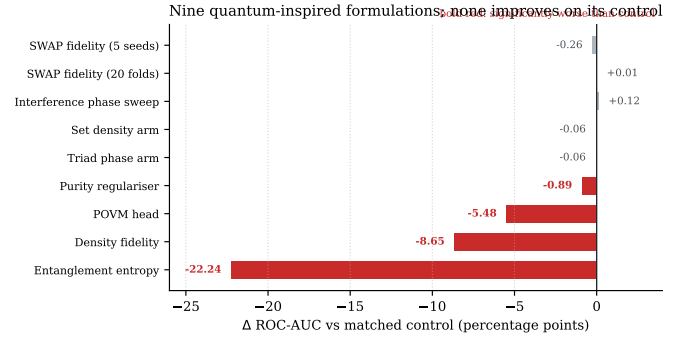


Fig. 6. Nine quantum-inspired formulations against classical controls matched for capacity or intent. None improves on its control; four are significantly worse. Bars are Δ ROC-AUC in percentage points.

C. Two further formulations

The set-level and triadic representations each admit a natural quantum reading, and we tested both as pre-registered controls.

A photo set is a mixed state $\rho = \frac{1}{N} \sum_i |\psi_i\rangle\langle\psi_i|$, and $\text{Tr}(\rho_1 \rho_2)$ is the corresponding fidelity. Added to the set features it changes ROC-AUC by -0.0006 , -0.0005 , $+0.0001$ and -0.0006 on the four benchmarks.

A child explained by two parents is naturally written as a coherent superposition with a relative phase, of which the classical convex mixture is the zero-phase special case. Sweeping the phase changes ROC-AUC by -0.0006 ($p = 0.147$).

This brings the total to nine formulations, none of which improves on its classical control.

D. Why the approach fails

The failures are structural rather than implementational. Interference-based formulations reduce to their classical counterparts because ℓ_2 -normalised FaceNet embeddings carry no phase structure to exploit; the phase sweep in Table VII is maximised

at the classical mixture. Density-matrix formulations require estimating a $d \times d$ operator from the images available for one identity — a median of five — so the resulting ρ is dominated by its leading eigenvector, which is the mean, while accumulating noise from tail directions that are not estimable. This is why $\text{Tr}(\rho_1 \rho_2)$ scores *below* simple mean-pooling.

E. An efficiency contribution

Independently of the above, we contribute an exact reformulation of the simulated SWAP test. The reference implementation materialises a 2^n statevector and permutes all n axes once per qubit, incurring kernel-launch overhead that leaves the accelerator idle. Applying rotations by reshaping to $(B, \text{left}, 2, \text{right})$ and precomputing the CNOT chain and R_z layer as a single diagonal phase vector yields a $380\times$ speedup on GPU ($192 \rightarrow 73,094$ pairs/s) with identical output and gradients, verified by test. We note that a closed-form product over qubits is *not* available here: the shared CNOT chain correlates the R_z phases, so the overlap does not factorise. The gain arises from removing overhead, not from approximation.

This is a quantum-versus-quantum speedup. Simulating a circuit is necessarily more costly than the classical operation it parallels, and we measure the quantum branch at $5\text{--}14\times$ slower than the classical path at every qubit count from 2 to 12.

IX. LIMITATIONS

Small test sets. KinFaceW-I and KinFaceW-II yield 212 and 400 test pairs per fold, giving confidence intervals of roughly ± 2 points. FIW and TSKinFace are firmer.

Single backbone. All results use frozen FaceNet embeddings. Whether the biases we quantify interact with backbone choice is untested.

Set-level gains are confined to one benchmark. Only FIW stores multiple photographs per identity, so the $+0.0771$ ROC-AUC of Section VII is measured on a single corpus; the other three are unchanged by construction. Whether the effect holds at other set sizes, and on corpora we could not test, is open.

Negative results are bounded. We tested nine formulations; we do not claim no quantum-inspired method can help on this task, only that these nine, covering the three natural roles, do not.

X. CONCLUSION

We have argued that facial kinship verification is conventionally posed and measured in ways that both understate and overstate what the data supports. Pair-level splitting leaks identities into the test set at a rate we measure at 100% on FIW; grouped k -fold with per-fold negative regeneration removes this and reduces seed-to-seed variation from 22.5 to 12.1 points. Against that protocol, treating verification as a person-level rather than photograph-level decision — standard in template-based face recognition, but not previously applied here — is worth $+0.077$ ROC-AUC on the one benchmark that supplies photograph sets, and by construction changes nothing on those that do not. Scoring parent–parent–child triads jointly adds $+0.032$ on TSKinFace.

We also report that nine quantum-inspired formulations, each against a capacity-matched classical control, fail to improve on their controls, with four significantly worse; the causes are structural rather than implementational. All code is released so that the protocol can be adopted and every claim checked.

Two limitations bound these results. Set-level gains are measured on a single corpus, because only FIW stores multiple photographs per identity. And cross-gender parent–child verification remains the weakest case: mother–son reaches 0.712 ROC-AUC on FIW against father–daughter’s 0.993, despite being the largest subset. We regard it as the most promising target for architectural work under the protocol proposed here.

REPRODUCIBILITY

Every numeric claim derives from JSON artefacts in `results/honest/`. The protocol is implemented in `src/splits.py`, `src/kfold.py`, and `src/ts_pairs.py`; results are regenerated by `scripts/evaluation/run_kfold.py`. The implementation is covered by 61 tests, including assertions that no group spans a fold boundary and that the fast SWAP-test reformulation matches the reference simulator in both value and gradient.

REFERENCES

- [1] L. Dawson, A. Zisserman, and C. Rupprecht, “Same photo: Cheating on visual kinship challenges,” in *Proc. British Machine Vision Conf. (BMVC)*, 2018.

- [2] C. Whitelam, E. Taborsky, A. Blanton, B. Maze, J. Adams, T. Miller, N. Kalka, A. K. Jain, J. A. Duncan, K. Allen *et al.*, “IARPA Janus Benchmark-B face dataset,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2017, pp. 90–98.
- [3] J. Lu, X. Zhou, Y.-P. Tan, Y. Shang, and J. Zhou, “Neighborhood repulsed metric learning for kinship verification,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 2, pp. 331–345, Feb. 2014.
- [4] S. Zhu, C. Gou, and F.-Y. Wang, “Kinship verification via relation-aware graph convolutional network,” *IEEE Trans. Multimedia*, vol. 24, pp. 4349–4361, 2022.
- [5] J. Lu, X. Zhou, Y.-P. Tan, Y. Shang, and J. Zhou, “Kinship verification in the wild: A benchmark,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2012, pp. 245–259.
- [6] X. Qin, X. Tan, and S. Chen, “Tri-subject kinship verification: Understanding the core of a family,” *IEEE Trans. Multimedia*, vol. 17, no. 10, pp. 1855–1867, Oct. 2015.
- [7] J. P. Robinson, M. Shao, Y. Wu, H. Wang, and Y. Fu, “Visual kinship recognition of families in the wild,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 11, pp. 2624–2637, Nov. 2018.
- [8] J. P. Robinson, M. Shao, and Y. Fu, “Recognizing families in the wild (RFIW): The 4th edition,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2020, pp. 858–866.
- [9] I. Kemelmacher-Shlizerman, S. M. Seitz, D. Miller, and E. Brossard, “The MegaFace benchmark: Efficient identification on a million scale,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 4873–4882.
- [10] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.
- [11] A. Torralba and A. A. Efros, “Unbiased look at dataset bias,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2011, pp. 1521–1528.
- [12] N. Kohli, D. Yadav, M. Vatsa, R. Singh, and A. Noore, “Supervised mixed norm metric learning for kinship verification on image sets,” *IEEE Trans. Inf. Forensics Security*, vol. 14, no. 5, pp. 1329–1342, 2019.
- [13] M. Schuld and N. Killoran, “Quantum machine learning in feature Hilbert spaces,” *Phys. Rev. Lett.*, vol. 122, no. 4, p. 040504, 2019.
- [14] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, “Supervised learning with quantum-enhanced feature spaces,” *Nature*, vol. 567, no. 7747, pp. 209–212, 2019.
- [15] E. Tang, “A quantum-inspired classical algorithm for recommendation systems,” in *Proc. 51st Annu. ACM SIGACT Symp. Theory of Comput. (STOC)*, 2019, pp. 217–228.